

MH2100 Calculus III

Midterm Practice 1 (Solutions)

AY 2026/2027, Semester 1

Quantitative Research Society @NTU

Practice material

Overview

- **Coverage:** curves and tangents; limits, continuity, partial derivatives and total differentiability; chain rule, directional derivatives, tangent lines, linear approximation and the Implicit Function Theorem; extrema and Lagrange multipliers; order reversal, polar coordinates and volume by a height double integral.
- **Practice design:** four original problems, 100 marks in total; intended for 110 minutes. This practice design is not a confirmed 2026 examination format.
- **Current administration:** Tuesday 6 October 2026, 16:30–18:20, venue TBA. One double-sided typed A4 sheet is permitted; attachments and sticky notes are not. No calculator policy is stated here.

Instructions

Answer **all** questions. Show all working clearly; answers without justification may receive zero marks.

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Question 1**(25 Marks)****PRACTICE PROBLEM 1.**

- (a) The curve C has parametrization $\mathbf{r}(t) = (t^2, t^3, 2t)$, $t \in \mathbb{R}$. Find the point on C at $t = 1$, a unit tangent vector there, and an equation of the tangent line. **(8 marks)**
- (b) Define $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$f(x, y) = \begin{cases} \frac{x^2 y}{x^4 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

Determine whether f is continuous at $(0, 0)$. Compute $f_x(0, 0)$ and $f_y(0, 0)$, and decide whether f is totally differentiable at $(0, 0)$. Justify every conclusion. **(17 marks)**

Solution**Method 1: Direct calculation**

(a)

$$\begin{aligned} \mathbf{r}(1) &= (1, 1, 2) \\ \mathbf{r}'(t) &= (2t, 3t^2, 2) \implies \mathbf{r}'(1) = (2, 3, 2) \\ \|\mathbf{r}'(1)\| &= \sqrt{17} \end{aligned}$$

$$\mathbf{T}(1) = \frac{1}{\sqrt{17}}(2, 3, 2), \quad (x, y, z) = (1, 1, 2) + s(2, 3, 2), \quad s \in \mathbb{R}.$$

(b)

$$\begin{aligned} f(x, x^2) &= \frac{x^4}{2x^4} = \frac{1}{2} \not\rightarrow 0 = f(0, 0) \quad (x \rightarrow 0, x \neq 0) && \text{(path } y = x^2) \\ f_x(0, 0) &= \lim_{h \rightarrow 0} \frac{f(h, 0)}{h} = 0, && f_y(0, 0) = \lim_{k \rightarrow 0} \frac{f(0, k)}{k} = 0 \\ \frac{|f(t, t^2)|}{\sqrt{t^2 + t^4}} &= \frac{1}{2|t|\sqrt{1 + t^2}} \not\rightarrow 0 && \text{(differentiable } \implies df = 0) \end{aligned}$$

$\therefore f$ not continuous $\implies$ not totally differentiable (differentiable $\implies$ continuous).

$$f \text{ is not continuous and is not totally differentiable at } (0, 0).$$

UNOFFICIAL QRS marking scheme (25 marks)

- (a) **(8 marks)** (i) **(2 marks)** point; (ii) **(3 marks)** derivative and normalization; (iii) **(3 marks)** tangent line.
- (b) **(17 marks)** (i) **(5 marks)** valid path and continuity conclusion; (ii) **(4 marks)** both partial derivatives; (iii) **(5 marks)** valid proof of non-differentiability, including the theorem that differentiability implies continuity; (iv) **(3 marks)** conclusion.

Question 2**(25 Marks)****PRACTICE PROBLEM 2.**

- (a) Let h be C^2 on a neighbourhood of $(1, 2)$, with $h(1, 2) = 3$ and $\nabla h(1, 2) = (4, -1)$. Define $G(x, y) = h(x^2 + y, xy + 2)$. At $P = (1, 0)$, find the directional derivative of G in the unit direction from P toward $(2, 2)$, the tangent line to $G(x, y) = 3$, and the linear approximation of $G(1.02, -0.03)$. **(15 marks)**
- (b) Let $F(x, y, z) = x^2 + ye^z + z^3 - 2$. Use the Implicit Function Theorem near $(1, 0, 1)$ to write $z = p(x, y)$. Find $p_x(1, 0)$ and $p_y(1, 0)$, then obtain the linear approximation of $p(1.04, -0.02)$. **(10 marks)**

Solution**Method 1: Chain rule and implicit differentiation**

- (a) $(u, v) = (x^2 + y, xy + 2) = (1, 2)$ at $P \implies (h_u, h_v) = (4, -1)$

$$\nabla G(P) = h_u(2x, 1) + h_v(y, x) = 4(2, 1) - (0, 1) = (8, 3) \quad (\text{chain rule})$$

$$\mathbf{u} = \frac{(2, 2) - (1, 0)}{\|(2, 2) - (1, 0)\|} = \frac{(1, 2)}{\sqrt{5}}$$

$$D_{\mathbf{u}}G(P) = \frac{(8, 3) \cdot (1, 2)}{\sqrt{5}} = \frac{14}{\sqrt{5}}.$$

$$G(P) = 3, \nabla G(P) = (8, 3) \implies \text{tangent line } 8(x - 1) + 3y = 0.$$

$$L(x, y) = 3 + 8(x - 1) + 3y \implies G(1.02, -0.03) \approx L(1.02, -0.03) = 3.07.$$

- (b) $F(1, 0, 1) = 0$, $F_z(1, 0, 1) = 3 \neq 0$, $F \in C^1 \implies z = p(x, y)$ near $(1, 0, 1)$ (IFT).

$$p_x = -\frac{F_x}{F_z} = -\frac{2}{3}, \quad p_y = -\frac{F_y}{F_z} = -\frac{e}{3} \quad (\text{at } (1, 0))$$

$$p(1.04, -0.02) \approx 1 - \frac{2}{3}(0.04) - \frac{e}{3}(-0.02) = 1 - \frac{2}{75} + \frac{e}{150} \approx 0.991455.$$

UNOFFICIAL QRS marking scheme (25 marks)

- (a) **(15 marks)**
- (i) **(5 marks)** chain-rule gradient;
 - (ii) **(4 marks)** unit direction and derivative;
 - (iii) **(3 marks)** tangent line;
 - (iv) **(3 marks)** linear approximation.
- (b) **(10 marks)**
- (i) **(3 marks)** verifies IFT hypotheses;
 - (ii) **(4 marks)** implicit derivatives;
 - (iii) **(3 marks)** approximation.

Question 3**(25 Marks)**

PRACTICE PROBLEM 3. Let $f(x, y) = xy$ on the closed elliptical disk $C = \{(x, y) \in \mathbb{R}^2 : x^2 + 4y^2 \leq 4\}$.

- (a) Find and classify every interior critical point of f . **(8 marks)**
- (b) Use Lagrange multipliers to find every extremum candidate of f on $x^2 + 4y^2 = 4$. **(12 marks)**
- (c) Establish that absolute extrema exist on C , then state the absolute maximum and minimum of f on C , including all points where they occur. **(5 marks)**

Solution**Method 1: Interior test and Lagrange multipliers**

- (a) $\nabla f = (y, x) = \mathbf{0} \iff (x, y) = (0, 0) \in \text{int } C$; $D = f_{xx}f_{yy} - f_{xy}^2 = 0 - 1 = -1 < 0$
(Second Derivative Test) $\implies (0, 0)$ is a saddle point.

- (b) $g = x^2 + 4y^2 \in C^1$, $\nabla g = (2x, 8y) \neq \mathbf{0}$ on $g = 4 \implies \nabla f = \lambda \nabla g$ (Lagrange):

$$\begin{aligned} y &= 2\lambda x, & x &= 8\lambda y, & x^2 + 4y^2 &= 4 \\ x = 0 \text{ or } y = 0 &\implies x = y = 0 \notin \{g = 4\}, \therefore xy \neq 0 \\ xy &= 16\lambda^2 xy \implies \lambda = \pm \frac{1}{4} \implies y = \pm \frac{x}{2} \\ x^2 + 4y^2 = 2x^2 = 4 &\implies x = \pm\sqrt{2}, & f = xy &= \pm\frac{x^2}{2} = \pm 1 \end{aligned}$$

$$\left(\sqrt{2}, \frac{\sqrt{2}}{2} \right), \left(-\sqrt{2}, -\frac{\sqrt{2}}{2} \right), \left(\sqrt{2}, -\frac{\sqrt{2}}{2} \right), \left(-\sqrt{2}, \frac{\sqrt{2}}{2} \right).$$

- (c) f continuous, C compact $\implies$ extrema exist (EVT); compare ± 1 on ∂C with $f(0, 0) = 0$:

$$\begin{aligned} \max_C f &= 1 \text{ at } \left(\sqrt{2}, \frac{\sqrt{2}}{2} \right), \left(-\sqrt{2}, -\frac{\sqrt{2}}{2} \right), \\ \min_C f &= -1 \text{ at } \left(\sqrt{2}, -\frac{\sqrt{2}}{2} \right), \left(-\sqrt{2}, \frac{\sqrt{2}}{2} \right). \end{aligned}$$

Method 2: AM–GM

$|xy| = \frac{1}{2}|x||2y| \leq \frac{1}{4}(x^2 + 4y^2) \leq 1$ on C (AM–GM), equality $\iff |x| = 2|y|$, $x^2 + 4y^2 = 4$
 $\therefore xy = \pm 1$ at $(\pm\sqrt{2}, \pm\frac{\sqrt{2}}{2})$: $+1$ for equal signs, -1 for opposite signs.

UNOFFICIAL QRS marking scheme (25 marks)

- (a) **(8 marks)** (i) **(3 marks)** solves $\nabla f = 0$; (ii) **(3 marks)** Hessian/discriminant; (iii) **(2 marks)** saddle conclusion.
- (b) **(12 marks)** (i) **(4 marks)** Lagrange system; (ii) **(4 marks)** all branches; (iii) **(4 marks)** candidates and values.
- (c) **(5 marks)** (i) **(2 marks)** continuity, compactness and EVT; (ii) **(3 marks)** global comparison.

Question 4**(25 Marks)****PRACTICE PROBLEM 4.**

- (a) Reverse the order of integration and evaluate

$$\int_0^1 \left[\int_{x^2}^x (1 + 2y) \, dy \right] dx.$$

(10 marks)

- (b) Let $D = \{(x, y) \in \mathbb{R}^2 : x \geq 0, y \geq 0, 1 \leq x^2 + y^2 \leq 4\}$. Find the volume of the solid above D and below $z = 6 - x^2 - y^2$. Use polar coordinates and a height double integral.
(15 marks)

Solution**Method 1: Reversal and polar height integral**

- (a)

$$\begin{aligned} R &= \{0 \leq x \leq 1, x^2 \leq y \leq x\} = \{0 \leq y \leq 1, y \leq x \leq \sqrt{y}\} \\ \iint_R (1 + 2y) \, dA &\stackrel{\text{Fubini}}{=} \int_0^1 \int_y^{\sqrt{y}} (1 + 2y) \, dx \, dy \\ &= \int_0^1 (\sqrt{y} - y)(1 + 2y) \, dy \\ &= \boxed{\left(\frac{2}{3} - \frac{1}{2}\right) + 2\left(\frac{2}{5} - \frac{1}{3}\right) = \frac{3}{10}.} \end{aligned}$$

- (b)

$$\begin{aligned} D : 0 \leq \theta \leq \frac{\pi}{2}, 1 \leq r \leq 2 & \quad (\text{polar}), \quad 6 - r^2 \geq 2 > 0 \text{ on } D \\ V &= \iint_D (6 - x^2 - y^2) \, dA = \int_0^{\pi/2} \int_1^2 (6 - r^2) r \, dr \, d\theta \quad (dA = r \, dr \, d\theta) \\ &= \frac{\pi}{2} \left[3r^2 - \frac{r^4}{4} \right]_1^2 = \boxed{\frac{21\pi}{8}} \end{aligned}$$

UNOFFICIAL QRS marking scheme (25 marks)

- (a)
- (10 marks)**

- (i) **(3 marks)** reversed region;
- (ii) **(3 marks)** bounds;
- (iii) **(2 marks)** inner integral;
- (iv) **(2 marks)** value $3/10$.

- (b)
- (15 marks)**

- (i) **(4 marks)** polar region;
- (ii) **(3 marks)** height and Jacobian;
- (iii) **(5 marks)** double integral;
- (iv) **(3 marks)** value $21\pi/8$. Equivalent correct Cartesian work receives full credit.